Audio Amplifier Design

Saikiran S
2024112007
saikiran.s@research.iiit.ac.in

Prasanna N R
2024102056
prasanna.r@students.iiit.ac.in

***Abstract*—The audio signal that is given as an input to the mic has peak-peak voltage of few milli-volts which is not sufficient to drive a speaker. This project deals with building a robust audio amplifier that amplifies the input signal and provides sufficient power to drive a speaker. This is done using an effective combination of a pre-amplifier, gain stage, band-pass filter and a power amplifier. The report includes concise calculations, simulation results of every stage.**

I. Introduction

The audio amplifier processes a low-level microphone input with a peak-to-peak amplitude of approximately 10–20 mV, which is inherently susceptible to noise and interference.

The first stage is a low-noise differential preamplifier that provides moderate voltage gain while offering strong common-mode noise rejection and input bias stability. This stage establishes a clean, balanced signal reference and suppresses interference picked up by the microphone and cabling.

The second stage is a common-emitter voltage amplifier with emitter degeneration, designed to provide the majority of the system's voltage gain. Emitter degeneration improves linearity, gain stability, and reduces harmonic distortion while maintaining adequate bandwidth.

The third stage implements an active band-pass filter using a Sallen–Key topology, restricting the signal bandwidth to the audio range (20 Hz – 20 kHz). This stage suppresses out-of-band noise and prevents unwanted frequency components from propagating into the power stage.

The final stage is a power amplifier configured to deliver sufficient current and power to drive a loudspeaker, while maintaining low output impedance and minimal distortion

Subsequent sections detail how each design specification is realized through circuit-level analysis and component selection. In cases where a specification cannot be fully met, the underlying limitations and trade-offs are explicitly discussed, along with the corresponding design rationale.

Specifications

- **Input Voltage to the Microphone:** $10–20\,\mathrm{mV_{pp}}$
- **Input Frequency Range:** $20\,\mathrm{Hz}–20\,\mathrm{kHz}$
- **Gain:** > 500 (achieved using pre-amplifier and gain stages only)
- **Power Output:** $> 1.5\,\mathrm{W}$
- **Load:** $\approx 8\,\Omega$

II. Pre Amplifier

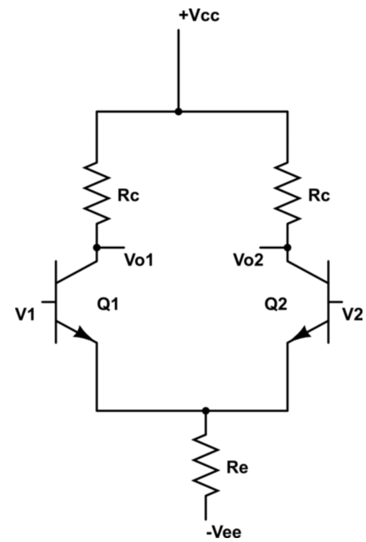

Fig. 1. Differential Amplifier

If V_{in1} and V_{in2} are the two input voltages of a differential amplifier, the output can be expressed as

$$V_{out} = A\,(V_{in1} - V_{in2}),$$

where A is the differential gain.

In the differential amplifier, the microphone signal is applied to one base terminal of the BJT pair, while the other base is grounded. The output is taken from the collector corresponding to the opposite input terminal, enabling differential operation and improved noise rejection.

Using DC analysis, the BJTs are biased to operate in the forward-active region, and the inherent symmetry of the differential amplifier is exploited to establish stable and balanced operating points.

The design involves two unknown parameters, R_C and R_E. Large-signal (DC) analysis is used to determine the biasing conditions, ensuring proper operation of the transistors, while small-signal analysis is employed to achieve the required voltage gain of 20. Using these analyses, appropriate values of R_C and R_E are selected.

A. *Large Signal Analysis*

Since the large-signal input is zero, the emitter node is biased at approximately $-0.7\,\text{V}$ to maintain the BJTs in forward-active mode, resulting in a collector current $I_C = 5\,\text{mA}$. With $\beta \approx 200$, the emitter current is approximately equal to the collector current, i.e., $I_E \approx I_C$. By symmetry, the total current flowing through the emitter resistor is $2I_C = 10\,\text{mA}$. Hence,

$$R_E = \frac{V_E - V_{EE}}{10\,\text{mA}} = \frac{4.3}{10\,\text{mA}} \approx 430\,\Omega.$$

. The transconductance of the BJT is given by

$$g_m = \frac{I_C}{V_T}.$$

For $I_C = 5\,\text{mA}$ and $V_T \approx 26\,\text{mV}$,

$$g_m = \frac{5 \times 10^{-3}}{26 \times 10^{-3}} \approx 0.192\,\text{S}.$$

The small-signal base–emitter resistance is

$$r_\pi = \frac{\beta}{g_m},$$

where $\beta = 200$. Hence,

$$r_\pi = \frac{200}{0.192} \approx 1.04\,\text{k}\Omega.$$

B. *Small Signal Analysis*

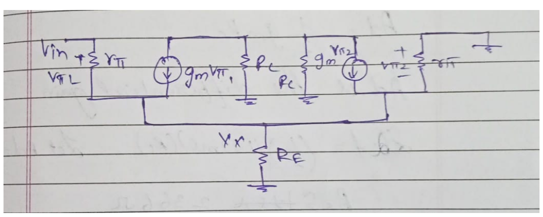

Fig. 2. Pre amp small signal

The small-signal node equation at the emitter node V_x is

$$V_x\left(\frac{1}{R_E} + \frac{1}{r_\pi}\right) = \frac{V_{in} - V_x}{r_\pi} + g_m V_{in} - 2g_m V_x.$$

Expanding and rearranging the terms,

$$V_x\left(\frac{1}{R_E} + \frac{2}{r_\pi} + 2g_m\right) = V_{in}\left(\frac{1}{r_\pi} + g_m\right).$$

Solving for V_x,

$$V_x = V_{in}\frac{\frac{1}{r_\pi} + g_m}{\frac{1}{R_E} + \frac{2}{r_\pi} + 2g_m}.$$

Hence, the emitter node gain is given by

$$\boxed{\frac{V_x}{V_{in}} = \frac{\frac{1}{r_\pi} + g_m}{\frac{1}{R_E} + \frac{2}{r_\pi} + 2g_m}}$$

The output voltage is taken at the collector and is given by

$$V_{out} = g_m R_C V_x.$$

Substituting the expression for V_x,

$$V_{out} = g_m R_C V_{in}\frac{\frac{1}{r_\pi} + g_m}{\frac{1}{R_E} + \frac{2}{r_\pi} + 2g_m}.$$

Hence, the overall small-signal voltage gain is

$$\boxed{\frac{V_{out}}{V_{in}} = g_m R_C\frac{\frac{1}{r_\pi} + g_m}{\frac{1}{R_E} + \frac{2}{r_\pi} + 2g_m}}$$

The differential gain after simplifying the above expression is given by

$$A_d = \frac{1}{2}g_m R_C.$$

We wanted a gain of 16.4, thus we calculate Rc Rearranging,

$$R_C = \frac{2A_d}{g_m}.$$

Substituting $A_d = 16.4$ and $g_m = 192\,\text{mS}$,

Thus, a standard resistor value of

$$\boxed{R_C = 175\,\Omega}$$

is chosen, resulting in a gain close to the desired specification. To determine the common-mode gain, the same input voltage is applied to both base terminals of the differential pair. Applying KCL at the emitter node gives

$$\frac{2(V_{in} - V_x)}{r_\pi} + 2g_m(V_{in} - V_x) = \frac{V_x}{R_E}.$$

Solving for V_x,

$$V_x = V_{in}\frac{2\left(\frac{1}{r_\pi} + g_m\right)}{\frac{1}{R_E} + 2\left(\frac{1}{r_\pi} + g_m\right)}.$$

The output voltage is given by

$$V_{out} = -g_m R_C(V_{in} - V_x).$$

Hence, the common-mode gain is

$$\boxed{A_c = \frac{V_{out}}{V_{in}} = -g_m R_C\left(1 - \frac{2\left(\frac{1}{r_\pi} + g_m\right)}{\frac{1}{R_E} + 2\left(\frac{1}{r_\pi} + g_m\right)}\right).}$$

Under the assumption that $g_m r_\pi \gg 1$ and R_E is sufficiently large, the above expression simplifies to

$$\boxed{A_c \approx -\frac{R_C}{2R_E}.}$$

C. CMRR calculations

Using $A_d \approx \frac{1}{2} g_m R_C$ and $A_c \approx -\frac{R_C}{2R_E}$, the ratio of differential gain to common-mode gain is

$$\boxed{\frac{A_d}{A_c} = g_m R_E.}$$

The common-mode rejection ratio is given by

$$\text{CMRR} = 20 \log_{10}\left(\frac{A_d}{A_c}\right).$$

With $\dfrac{A_d}{A_c} = 83$,

$$\text{CMRR} = 20 \log_{10}(83) \approx 20 \times 1.919 \approx 38.4\,\text{dB}.$$

D. Output Resistance Calculation

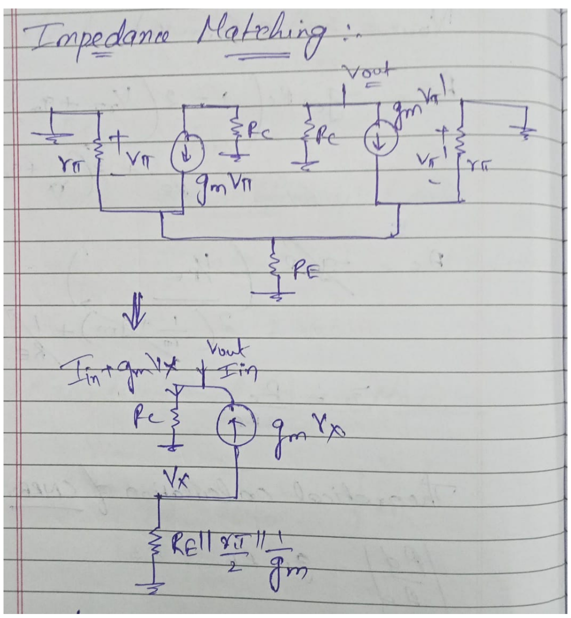

Fig. 3. Differential Amplifier

To determine the output resistance, the input is set to zero and a test current I_{in} is applied at the output node.

From small-signal analysis,

$$\frac{V_x}{R_E \parallel \frac{r_\pi}{2} \parallel \frac{1}{g_m}} = -g_m V_x$$

This implies

$$V_x = 0$$

Hence, the output voltage becomes

$$V_{out} = I_{in} R_C$$

Therefore,

$$\frac{V_{out}}{I_{in}} = R_C$$

$$R_{out} = R_C$$

Substituting the calculated value of R_C,

$$\boxed{R_{out} = 236.6\,\Omega}$$

Thus, to avoid attenuation of the signal due to loading effects, the input impedance of the gain stage must be significantly larger than the output resistance of the previous stage.

As a standard design rule,

$$R_{in,\text{gain}} \geq 10\, R_{out}$$

This ensures minimal voltage division between the stages and preserves the intended signal amplitude.

III. Gain Stage

The gain stage is a common-emitter amplifier with emitter degeneration.

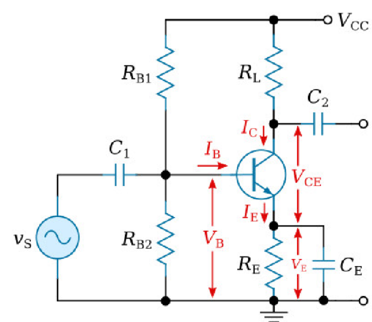

Fig. 4. Gain stage

A. Large Signal Analysis

For designing the second stage, the following assumptions are made:

1) $A_v = 22.5$
2) $V_{BE} = 0.65\,\text{V}$
3) $V_{O(\text{DC})} = 0.9\,\text{V}$

We choose these three assumptions for designing as :
The output should have the maximum voltage swing. Clipping was observed in simulations when $V_{O(DC)}$ was taken to be 0.65-0.7 (Theoretically this provides the maximum swing while maintaining the BJT in forward active mode). Thus a higher $V_{O(DC)}$ was chosen and calculations done based on this.

DC and Small-Signal Parameters:

$$I_C = I_S e^{\frac{V_{BE}}{V_T}} = 0.72\,\text{mA}$$

$$R_C = \frac{V_{CC} - V_O}{I_C} = \frac{5 - 0.9}{0.72\,\text{mA}} \approx 5.7\,\text{k}\Omega$$

$$g_m = \frac{I_C}{V_T} = \frac{0.72\,\text{mA}}{26\,\text{mV}} = 27.6\,\text{mS}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{200}{27.6\,\text{mS}} \approx 7.24\,\text{k}\Omega$$

The base voltage is

$$V_B = V_{BE} + V_E = 0.65 + (-4.86) = -4.2\,\text{V}$$

Applying KVL across the bias divider from $+5\,\text{V}$ to $-5\,\text{V}$,

$$\frac{5 + 4.2}{R_{B1}} = \frac{0.8}{R_{B2}}$$

$$\frac{R_{B1}}{R_{B2}} = 11.5$$

To ensure stiff biasing, the divider current is chosen as

$$I_{div} = 10 I_B = 10 \times 3.6\,\mu\text{A} = 36\,\mu\text{A}$$

Thus,

$$R_{B1} + R_{B2} = \frac{10}{36\,\mu\text{A}} \approx 138.8\,\text{k}\Omega$$

Using

$$R_{B1} = 11.5 R_{B2}$$

$$11.5 R_{B2} + R_{B2} = 138.8\,\text{k}\Omega$$

$$12.5 R_{B2} = 138.8\,\text{k}\Omega$$

$$R_{B2} \approx 11.1\,\text{k}\Omega$$

$$R_{B1} \approx 127.7\,\text{k}\Omega$$

$$\boxed{R_{B1} \approx 128\,\text{k}\Omega, \quad R_{B2} \approx 11\,\text{k}\Omega}$$

B. Small Signal Calculations

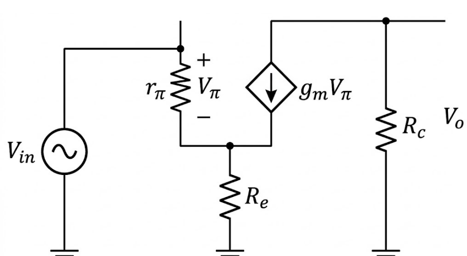

Fig. 5. Small Model

Voltage Gain Derivation: Applying KCL at node V_x,

$$\frac{V_x}{R_E} = \frac{V_{in} - V_x}{r_\pi} + g_m(V_{in} - V_x)$$

Rearranging,

$$\frac{V_x}{R_E} = \left(\frac{1}{r_\pi} + g_m\right)(V_{in} - V_x)$$

$$V_x = R_E\left(\frac{1}{r_\pi} + g_m\right)(V_{in} - V_x)$$

$$V_x\left[1 + R_E\left(\frac{1}{r_\pi} + g_m\right)\right] = R_E\left(\frac{1}{r_\pi} + g_m\right)V_{in}$$

$$\frac{V_x}{V_{in}} = \frac{R_E\left(\frac{1}{r_\pi} + g_m\right)}{1 + R_E\left(\frac{1}{r_\pi} + g_m\right)}$$

The output voltage is

$$V_o = -g_m R_C (V_{in} - V_x)$$

Substituting,

$$\frac{V_o}{V_{in}} = -g_m R_C \frac{1}{1 + R_E\left(\frac{1}{r_\pi} + g_m\right)}$$

If $\frac{1}{r_\pi} \ll g_m$, then

$$A_v = \frac{V_o}{V_{in}} \approx -\frac{g_m R_C}{1 + g_m R_E}$$

Since the first stage provides a gain of approximately 16.5, the second stage is designed to provide

$$A_v = -22.5$$

Using the previously derived expression,

$$A_v \approx -\frac{g_m R_C}{1 + g_m R_E}$$

Substituting $g_m = 27.6\,\text{mS}$ and $R_C = 5.7\,\text{k}\Omega$,

$$22.5 = \frac{(27.6 \times 10^{-3})(5.7 \times 10^3)}{1 + (27.6 \times 10^{-3})R_E}$$

$$22.5 = \frac{157.32}{1 + 0.0276 R_E}$$

Solving,

$$1 + 0.0276 R_E = \frac{157.32}{22.5}$$

$$1 + 0.0276 R_E \approx 6.99$$

$$0.0276 R_E \approx 5.99$$

$$R_E \approx 190\,\Omega$$

$$\boxed{R_E = 190\,\Omega}$$

Verification of Forward-Active Operation: For a BJT to operate in the forward-active region,

$$V_{BE} > 0 \quad \text{and} \quad V_{BC} < 0$$

The collector voltage is

$$V_C = V_{CC} - I_C R_C$$

$$V_C = 5 - (0.72\,\text{mA})(5.7\,\text{k}\Omega)$$

$$V_C = 5 - 4.10 = 0.9\,\text{V}$$

The base voltage is

$$V_B = -4.2\,\text{V}$$

Thus,

$$V_{BC} = V_B - V_C$$

$$V_{BC} = -4.2 - 0.9$$

$$V_{BC} = -5.1\,\text{V}$$

Since

$$V_{BC} < 0$$

the collector–base junction is reverse biased.

$$\boxed{\text{Hence, the BJT operates in the forward-active region.}}$$

C. Rin derivation

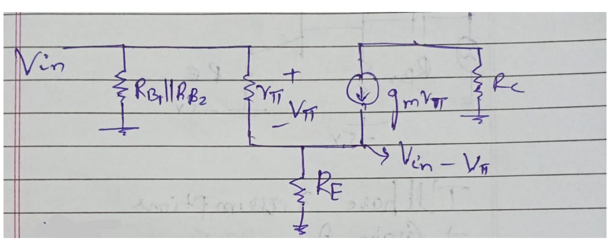

Fig. 6. Small Model

Applying KCL at the base–emitter small-signal model,

$$\frac{V_\pi}{r_\pi} + g_m V_\pi = \frac{V_{in} - V_\pi}{R_E}$$

Rearranging,

$$V_\pi \left(\frac{1}{r_\pi} + g_m + \frac{1}{R_E} \right) = \frac{V_{in}}{R_E}$$

$$\frac{V_\pi}{V_{in}} = \frac{\dfrac{1}{R_E}}{\dfrac{1}{r_\pi} + g_m + \dfrac{1}{R_E}}$$

The total input current is

$$I_{in} = \frac{V_{in}}{R_{B1} \parallel R_{B2}} + \frac{V_\pi}{r_\pi}$$

Substituting for V_π,

$$I_{in} = \frac{V_{in}}{R_{B1} \parallel R_{B2}} + \frac{V_{in}}{r_\pi} \cdot \frac{\dfrac{1}{R_E}}{\dfrac{1}{r_\pi} + g_m + \dfrac{1}{R_E}}$$

Hence,

$$R_{in} = \frac{V_{in}}{I_{in}} = \left[\frac{1}{R_{B1} \parallel R_{B2}} + \frac{1}{r_\pi} \cdot \frac{\dfrac{1}{R_E}}{\dfrac{1}{r_\pi} + g_m + \dfrac{1}{R_E}} \right]^{-1}$$

Substituting

$$r_\pi = 7.24\,\text{k}\Omega, \quad g_m = 27.6\,\text{mS}, \quad R_E = 190\,\Omega, \quad R_{B1} \parallel R_{B2} \approx 10.1\,\text{k}\Omega$$

$$\boxed{R_{in} \approx 8.26\,\text{k}\Omega}$$

Loading Verification: From the previous stage, the output resistance was

$$R_{out} = 236.6\,\Omega$$

To avoid attenuation due to loading,

$$R_{in,\text{min}} = 10\,R_{out} = 10 \times 236.6 = 2.366\,\text{k}\Omega$$

Since the calculated input resistance of the gain stage is

$$R_{in} = 8.26\,\text{k}\Omega$$

$$\boxed{R_{in} > R_{in,\text{min}}}$$

Thus, the loading effect is negligible and the inter-stage signal attenuation is avoided.

IV. Filter

The filter stage is implemented using cascaded Sallen–Key high-pass and low-pass sections to realize a band-pass response over 20 Hz – 20 kHz.

A standard second-order filter provides a roll-off of 20 dB/decade. Since both sections are second-order, cascading them results in a fourth-order band-pass filter.

Hence, the overall roll off outside the passband is 40 dB/decade, providing sharper cutoff and improved noise rejection.

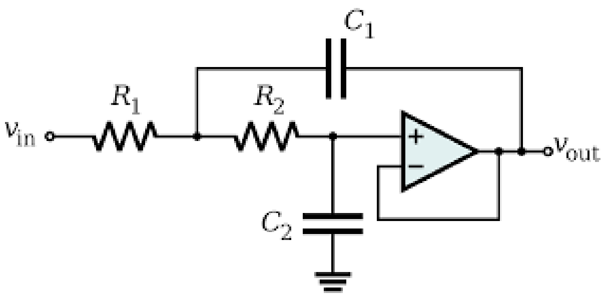

Fig. 7. Low pass filter

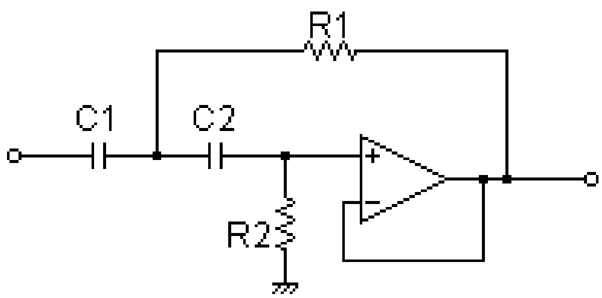

Fig. 8. High pass Filter

A. Low-Pass Filter Design

Let $V_x(s)$ be the intermediate node voltage between R_1 and R_2.

Applying KCL at $V_x(s)$,

$$\frac{V_x - V_{in}}{R_1} + \frac{V_x - V_o}{R_2} + \frac{V_x - V_o}{X_{C1}} = 0$$

The output is given by

$$V_o(s) = \frac{X_{C2}}{X_{C2} + R_2} V_x(s)$$

After simplification, the transfer function becomes

$$H(s) = \frac{V_o(s)}{V_{in}(s)} = \frac{1}{1 + sC_2(R_1 + R_2) + s^2 C_1 C_2 R_1 R_2}$$

A standard second-order low-pass filter is of the form

$$H(s) = \frac{\omega_0^2}{s^2 + 2\alpha s + \omega_0^2}$$

where

$$\omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}, \qquad 2\alpha = \frac{R_1 + R_2}{R_1 R_2 C_1}$$

$$Q = \frac{\omega_0}{2\alpha}$$

Let

$$R_1 = mR, \quad R_2 = \frac{R}{m}, \quad C_1 = nC, \quad C_2 = \frac{C}{n}$$

Then

$$Q = \frac{mn}{m^2 + 1}$$

For

$$Q = \frac{1}{2}, \quad m = n = 1$$

Choosing

$$f_0 = 50\,\text{kHz}$$
$$R_1 = \frac{1}{2\pi f_0 C_1}$$

Let

$$C_1 = C_2 = 0.1\,\text{nF}$$

$$R_1 = R_2 = \frac{1}{2\pi(50 \times 10^3)(0.1 \times 10^{-9})} \approx 27\,\text{k}\Omega$$

B. High-Pass Filter Design

Let $V_x(s)$ be the node voltage between C_1 and C_2. Applying KCL at $V_x(s)$,

$$\frac{V_x - V_{in}}{X_{C1}} + \frac{V_x - V_{out}}{R_1} + \frac{V_x - V_{out}}{X_{C2}} = 0$$

The output voltage is

$$V_{out}(s) = V_x(s)\left(\frac{R_2}{R_2 + X_{C2}}\right)$$

After simplification, the transfer function becomes

$$H(s) = \frac{s^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

where

$$\omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}, \qquad \frac{\omega_0}{Q} = \frac{C_1 + C_2}{R_2 C_1 C_2}$$

Let

$$R_1 = mR, \quad R_2 = \frac{R}{m}, \quad C_1 = \frac{C}{n}, \quad C_2 = nC$$

Then

$$Q = \frac{n}{m(n^2 + 1)}$$

Choosing

$$m = 0.1, \quad Q = 0.7$$

$$0.0707n^2 - 0.1n + 0.0707 = 0$$

$$n \approx 14.07 \quad \text{or} \quad n \approx 0.07$$

The cutoff frequency is

$$f_0 = \frac{1}{2\pi RC}$$

We will have

$$f_0 = 10Hz$$

Let

$$R_1 = 16\,\text{k}\Omega, \quad R_2 = 1.6\,\text{M}\Omega$$

$$f_0 = \frac{1}{2\pi C(16 \times 10^3)}$$

Choosing

$$C = 0.1\,\mu\text{F}$$

$$C_1 = \frac{C}{n} = 7.2\,\text{nF}, \qquad C_2 = nC = 1.4\,\mu\text{F}$$

V. Power Amplifier

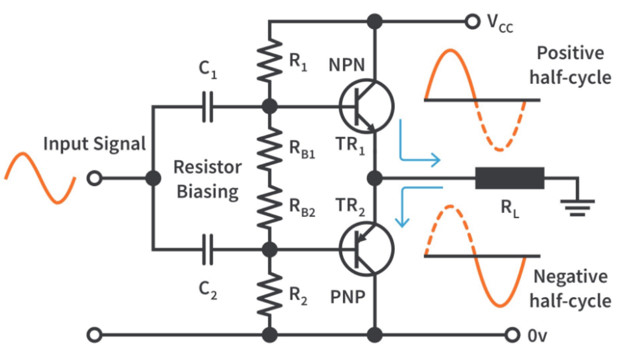

Fig. 9. Class AB Power-Amp

We design the power amp such that both the NPN and the PNP power transistors are biased near cutoff. On doing so the NPN allows the positive half cycle of the input waveform (as the PNP will be in cutoff) and the PNP allows the negative half cycle of the input waveform(as the NPN will be in cutoff) D

A. Biasing and Small-Signal Analysis of Output Stage

By symmetry,

$$R_1 = R_2 = R = 1k\Omega$$

The diodes are designed to have a drop of

$$V_{BE} = 0.65\,\text{V}$$

with non-ideality factor

$$\eta = 1.75$$

and

$$I_S = 2.52\,\text{nA}$$

Using the diode equation,

$$I_C = I_S e^{\frac{V_{BE}}{\eta V_T}} = 3.5\,\text{mA}$$

The base current is

$$I_B = 0.6\,\text{mA}$$

Hence, the current through the bias resistor is

$$I_R = 4.1\,\text{mA}$$

$$R = \frac{5 - 0.65}{4.1\,\text{mA}} = \frac{4.35}{4.1\,\text{mA}} \approx 1.03\,\text{k}\Omega$$

The transconductance is

$$g_m = \frac{dI_C}{dV_{BE}} \approx \frac{I_C}{V_T}$$

Since

$$V_{out} = I_E R_L \approx I_C R_L$$

and

$$V_{in} - V_{BE} = V_{out}$$

we get

$$V_{out} = V_{in} - \frac{I_C}{g_m}$$

Noting that

$$\frac{1}{g_m} = r_e$$

$$V_{out} = V_{in} - I_C r_e$$

$$I_C R_L = V_{in} - I_C r_e$$

$$V_{in} = I_C(R_L + r_e)$$

$$\frac{V_{out}}{V_{in}} = \frac{R_L}{R_L + r_e}$$

Since typically

$$R_L \gg r_e$$

$$\frac{V_{out}}{V_{in}} \approx \frac{R_L}{R_L} = 1$$

Thus, the voltage gain of the emitter follower output stage is approximately unity.The output of the power amplifier for an 8Ω speaker is :

$$P_{AC} = \frac{V_{out-rms}^2}{R_L} \approx \frac{\frac{8.8}{2} * \frac{8.8}{2}}{2 * 8} \approx 1.21W$$

VI. LTSPICE SIMULATIONS

A. Pre Amplifier

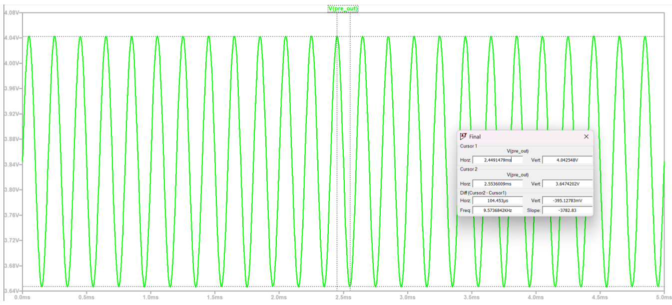

Fig. 10. Transient Response for 10k and 20mVpp.

B. Gain contribution from gain stage

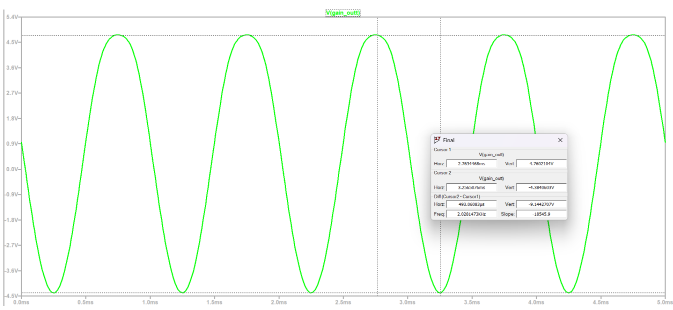

Fig. 11. The input signal is applied to the gain stage.

The gain of the gain stage is

$$A_v = \frac{457}{20} \approx 22.85$$

C. Gain Stage and Pre Amp Integration

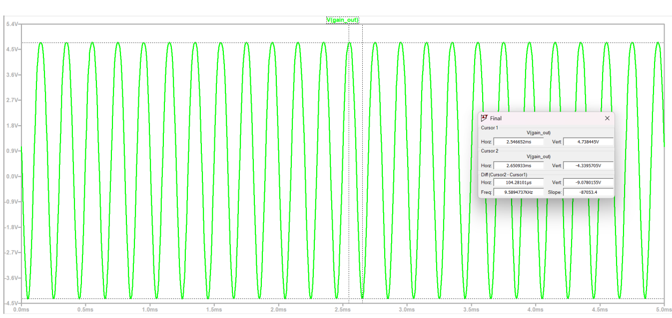

Fig. 12. Transient Response for 10k and 20mVpp.

D. Filter Stage

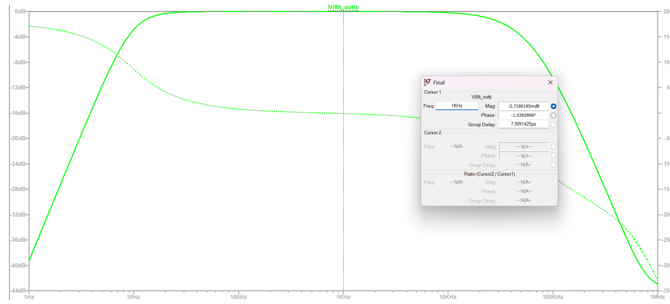

Fig. 13. Frequency response of bandpass filter.

It is observed from the frequency response that the mid-band gain of the filter stage is approximately 0 dB, indicating unity gain within the passband.

E. Power Amplifier

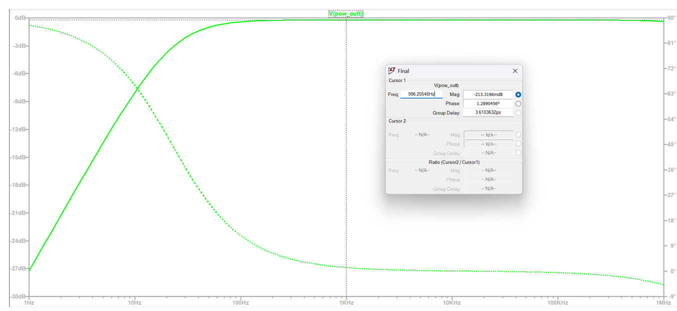

Fig. 14. Frequency response of Power Amplifier.

It is observed from the frequency response that the mid-band gain of the Power Amplifier is approximately 0 dB.

F. Final

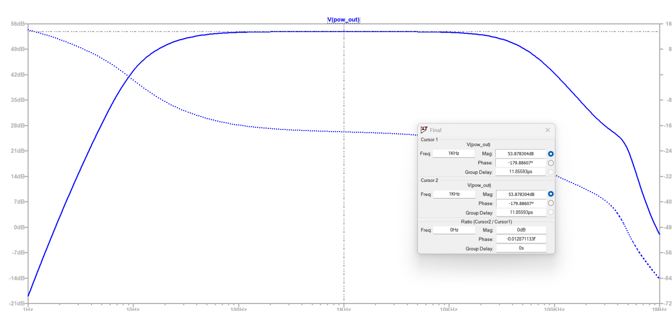

Fig. 15. Frequency Response of the final circuit.

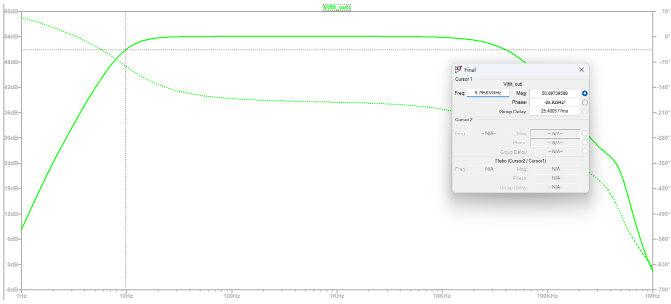

Fig. 16. The lower cutoff frequency is $f_l = 9.79\,\text{Hz}$..

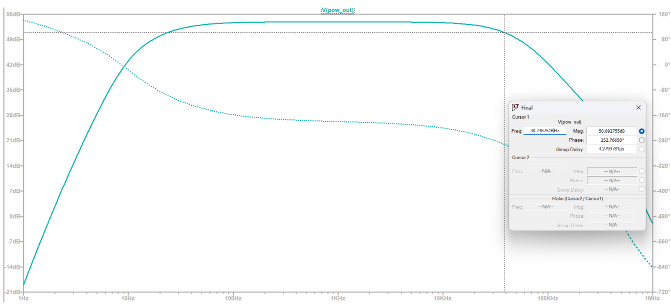

Fig. 17. The upper cutoff frequency is $f_h = 38.74\,\text{kHz}$..

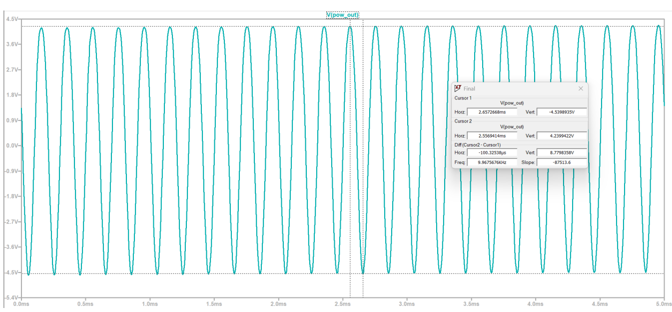

Fig. 18. Transient Response of the final circuit after power amp.

VII. Hardware Results

A. Pre Amplifier

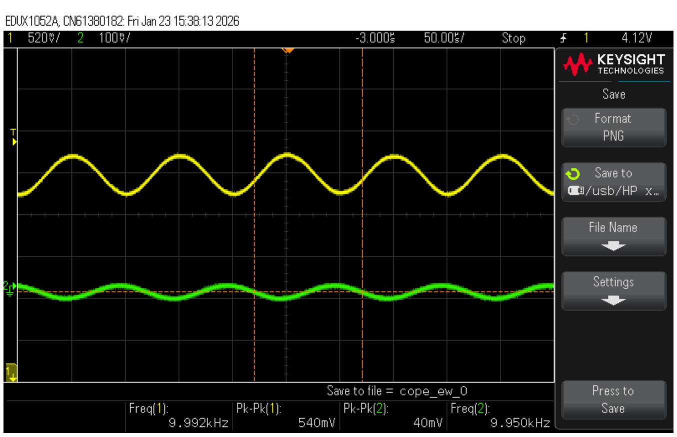

Fig. 19. Transient Response of the Pre Amplifer.

B. Pre amp stage and Gain Stage integration

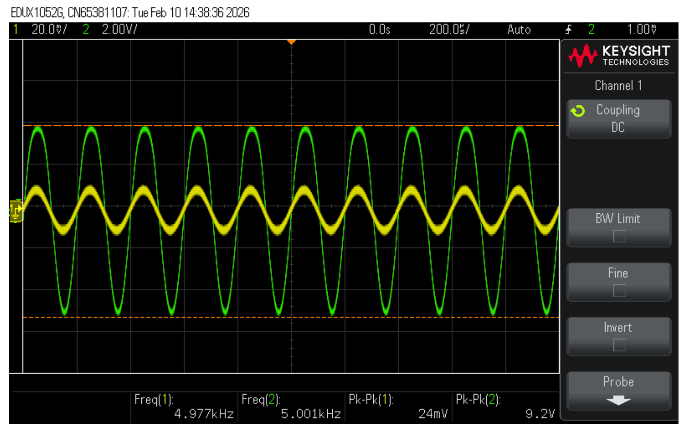

Fig. 20. Transient Response of the integrated stages.

C. Filter Stage

D. Lowpass Filter

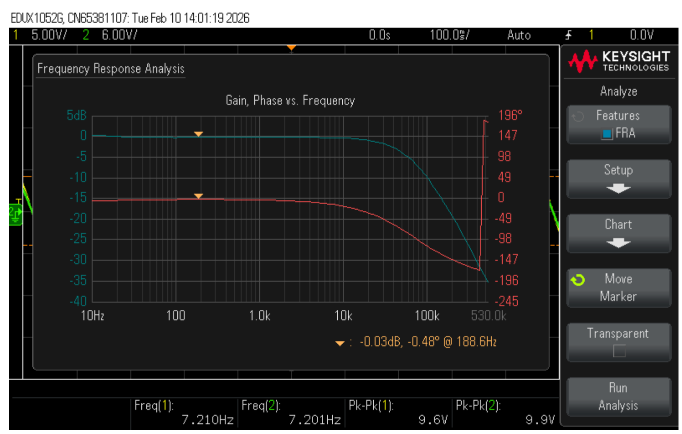

Fig. 21. Frequency response of Low pass filter.

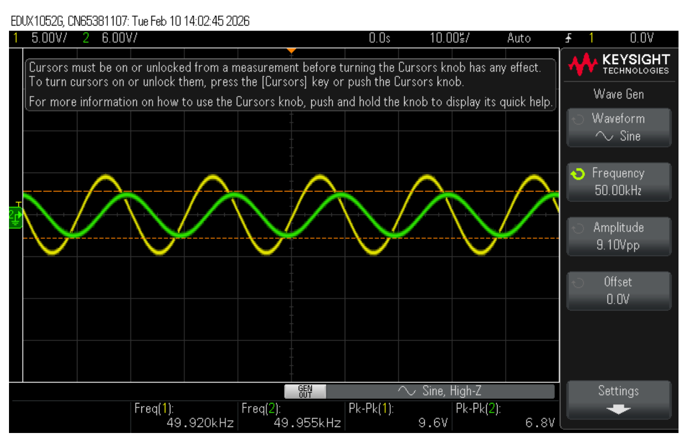

Fig. 22. Filter output attenuating at f=50KHz.

E. Highpass Filter

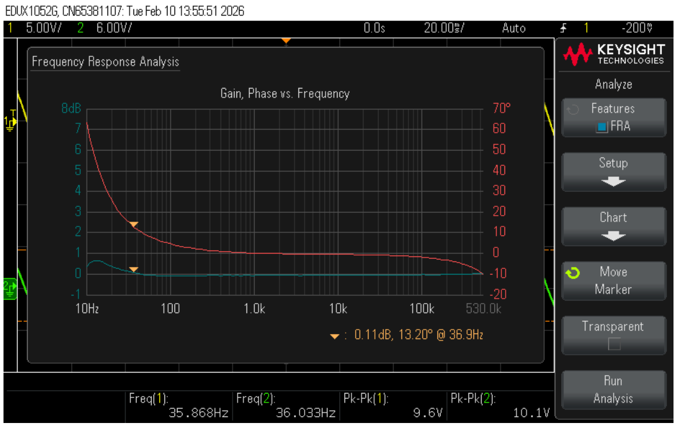

Fig. 23. Frequency response of High pass Filter.

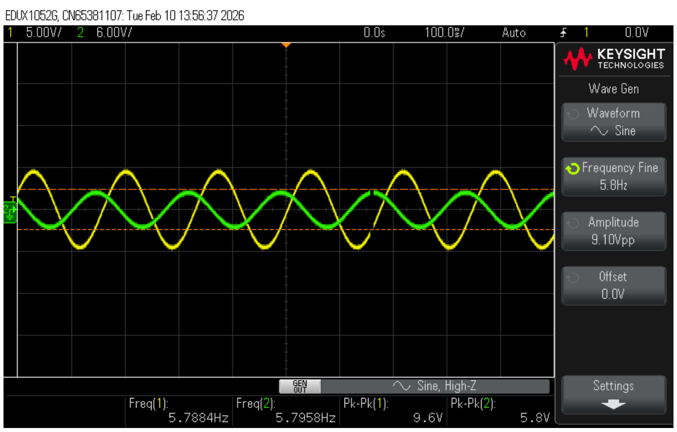

Fig. 24. Filter output attenuating at f=5.8Hz.

F. Final

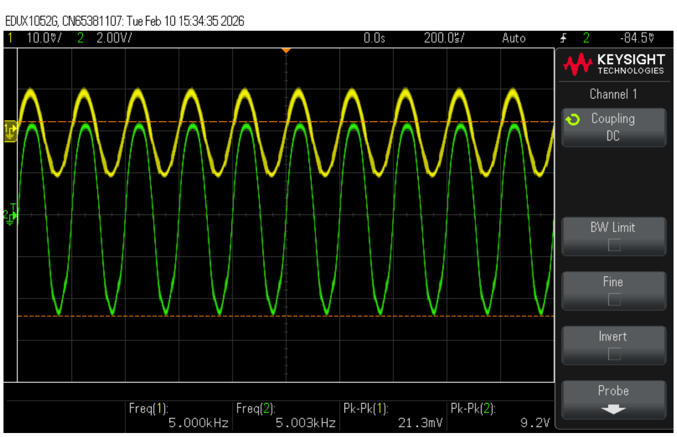

Fig. 25. Transient Response of the final circuit.

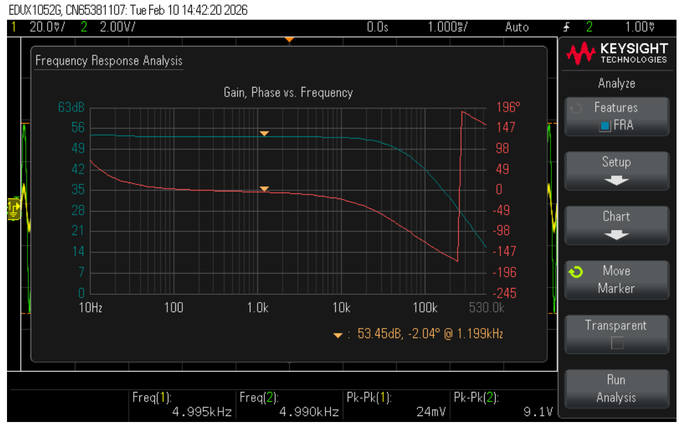

Fig. 26. Frequency Response of the final circuit.

VIII. Circuit Parameters

A. CMRR

The differential gain is measured as

$$A_d = 17$$

The common-mode gain is obtained by applying the same input to both terminals of the differential pair and measuring the output:

$$A_c = \frac{5.3}{20} = 0.265$$

The Common-Mode Rejection Ratio (CMRR) is then calculated as

$$\text{CMRR} = 20\log_{10}\left(\frac{17}{0.265}\right)$$

$$\boxed{\text{CMRR} \approx 36.1\,\text{dB}}$$

B. Total Harmonic Distortion (THD)

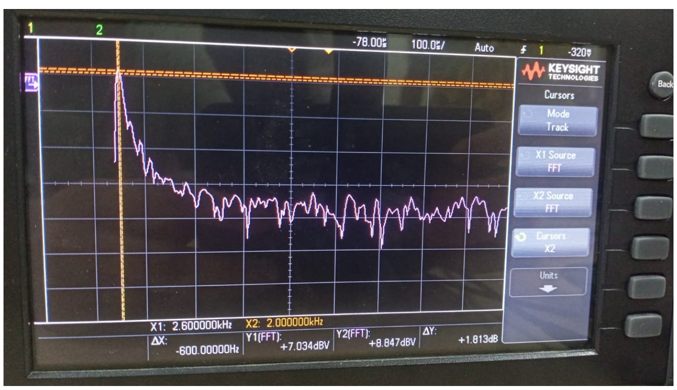

Fig. 27. FFT plot for f=2KHz.

The total harmonic distortion is calculated using

$$THD = \frac{\sqrt{\sum_{n=2}^{\infty} V_{n,\text{RMS}}^2}}{V_{f,\text{RMS}}}$$

The worst-case distortion values observed were:

$$THD = 8.09\% \quad \text{at } 10\,\text{mV},\ 20\,\text{kHz}$$

$$THD = 9.08\% \quad \text{at } 10\,\text{mV},\ 20\,\text{Hz}$$

The best-case distortion measured was

$$\boxed{THD = 2.56\%}$$

C. Slew Rate

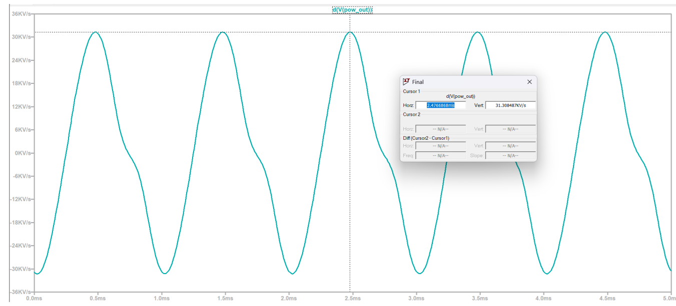

Fig. 28. Measuring slew rate of the circuit.

The slew rate turns out to be 31.3KV/s. We observe a linear rise in voltage output due to the slew rate of the op-amp not being able to track the input voltage accurately. Hence while implementing the hardware we choose an op-amp with a better slew rate - TL081.

D. AC Power Calculation

The AC output power delivered to the load is given by

$$P_{AC} = \frac{V_{\text{RMS}}^2}{R_{load}} = \frac{V_{\text{peak}}^2}{2R_{load}}$$

Substituting,

$$P_{AC} = \frac{4.6^2}{2 \times 10}$$

$$P_{AC} = \frac{21.16}{20}$$

$$\boxed{P_{AC} \approx 1.06\,\text{W}}$$

IX. Conclusion

The designed audio amplifier achieves a gain of over 450 with acceptable distortion and bandwidth. We have made key improvements such as implementing higher-order filters that has sharper Q-factor and higher slope. However few more improvements Thermal compensation techniques through Thermal compensation techniques in the Class-AB output stage to reduce bias drift and crossover distortion, Integration of negative feedback across stages to reduce THD and improve linearity and others.

References

[1] Wikipedia, “Sallen–Key Topology.”
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[3] B. Razavi, “Common-Emitter (CE) Amplifier,” YouTube Lecture.
[4] J. Fiore, “Electronics Lecture Series,” YouTube.